



# **ANALYSIS OF THE 2004-05 ENGLISH PREMIER LEAGUE SEASON**

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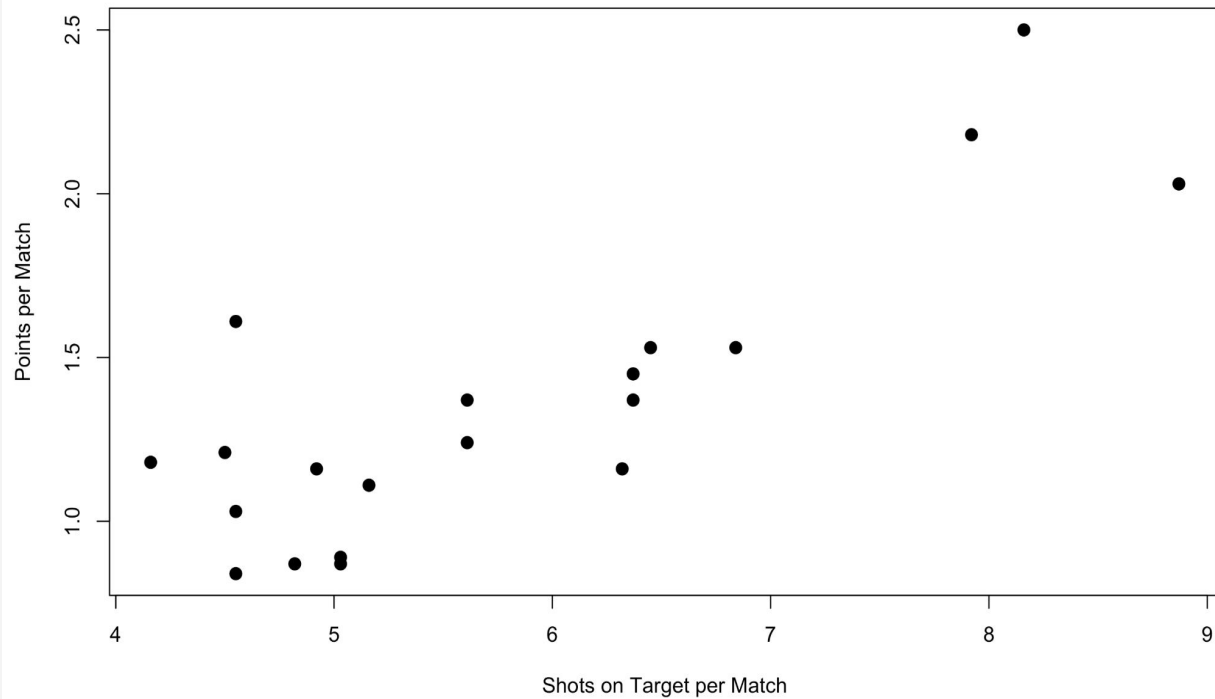
By: Chelsea Seegars

# SEASON OVERVIEW + FORMAT

Chelsea Football Club who just hired their new manager Jose Mourinho ended Arsenal's reign as Premier League champions. They broke the record for fewest goals conceded in Premier League history with 15 goals conceded and the record for most clean sheets in a season for a goalkeeper with Petr Čech having 24 clean sheets that season. Chelsea secured the title three weeks early and were known for their dominant defense. Will Chelsea be just as dominant in the simulated model? Crystal Palace, Norwich City and Southampton were all relegated by finishing with less than the classic 40 point threshold. Will the simulated model be more favorable to Crystal Palace whose fight to stay in the top division ended on the final day?

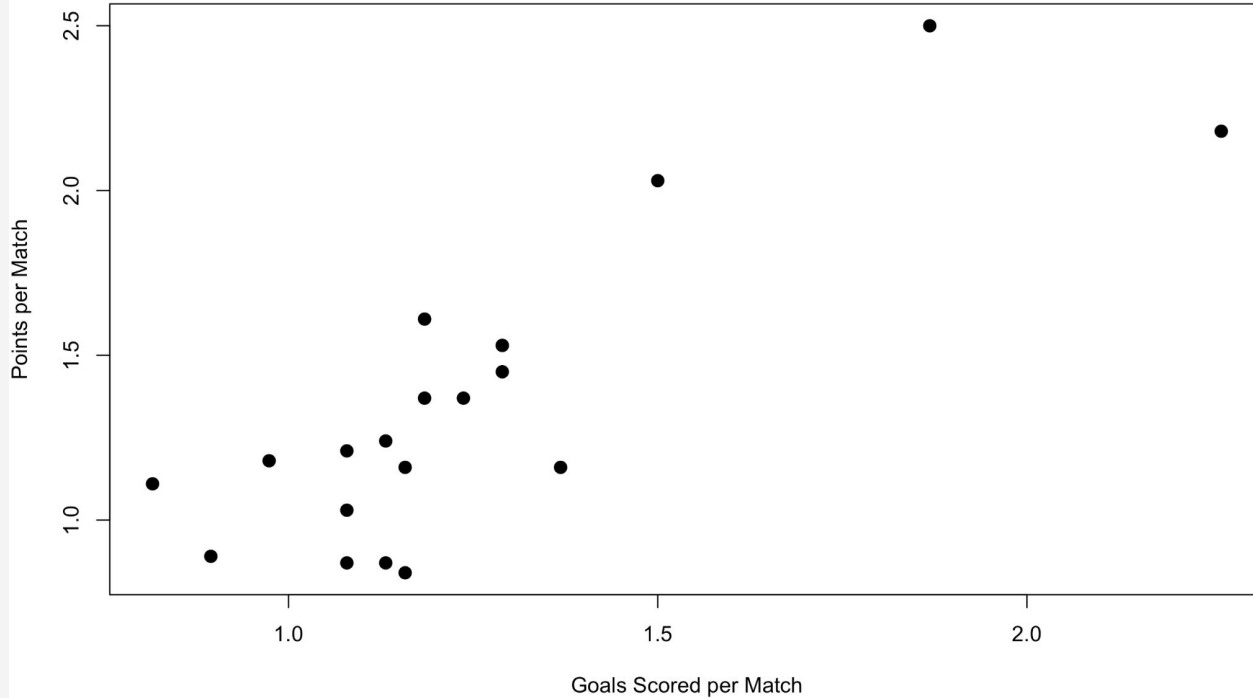
The English Premier League is composed of 20 teams, the bottom three teams are relegated to the English Championship and the top four teams compete in the UEFA Champions League. All 20 teams compete in the FA Cup and EFL Cup (now Carabao Cup) during the Premier League Season too.





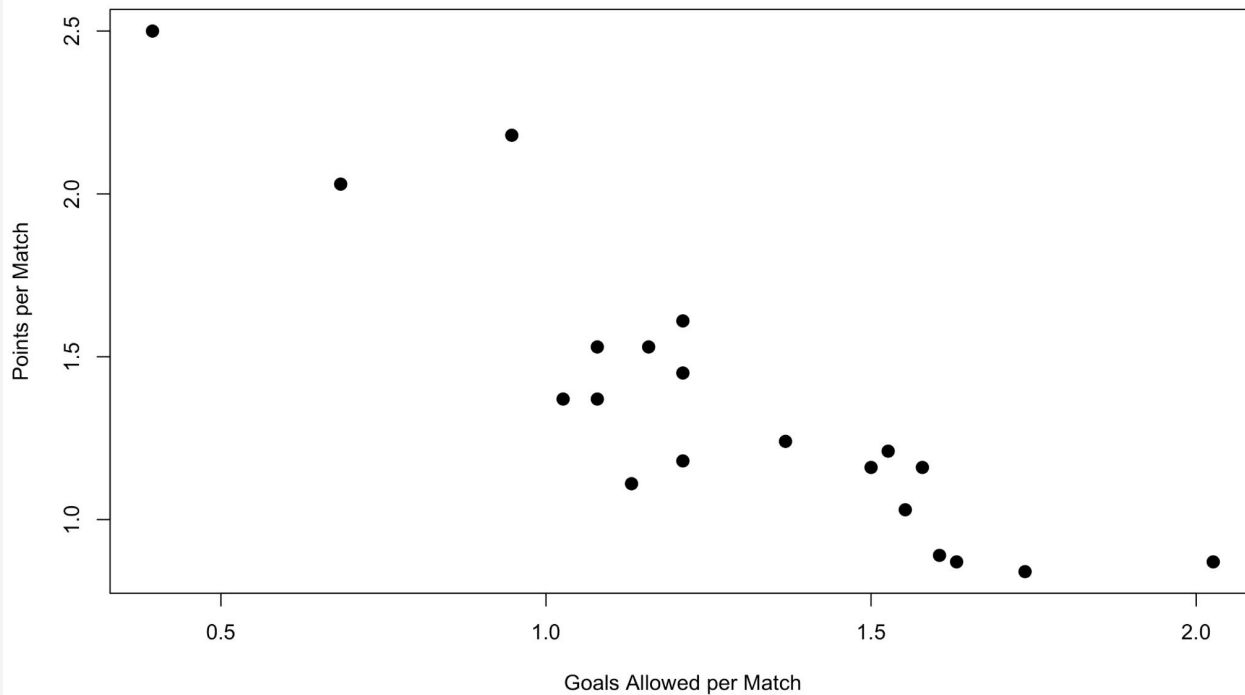
**CORRELATION**  
**.8302804**

**SHOTS ON TARGET PER MATCH V. POINTS PER MATCH**



**CORRELATION**  
**0.8283142**

**GOALS SCORED PER MATCH V. POINTS PER MATCH**

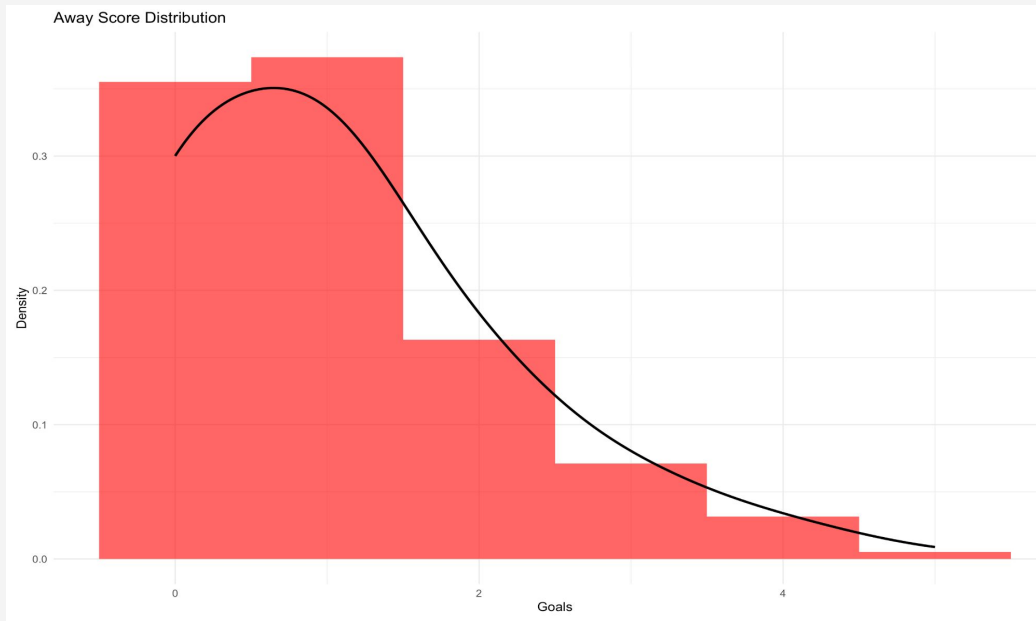


**CORRELATION**  
**-0.8920112**

**GOALS ALLOWED PER MATCH V. POINTS PER MATCH**

# DISCUSSION OF RELEVANT STATISTICS

All three graphs show strong linear relationships between points per match and goals scored per match, points per match and goals allowed per match and points per match and shots on target per match. Shots on target per match and goals scored per match had strong positive linear relationships which makes sense since the more shots you take on target the more likely one of them will convert into a goal and the more goals that are scored the more likely the team is to win (implying they get more points). Goals allowed per match has a strong negative correlation which also makes sense since the more goals a team concedes the more likely they are to lose (implying they get less points).

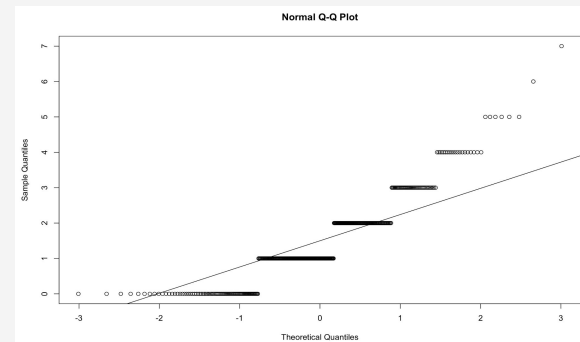


## **SHAPIRO-WILKS TEST**

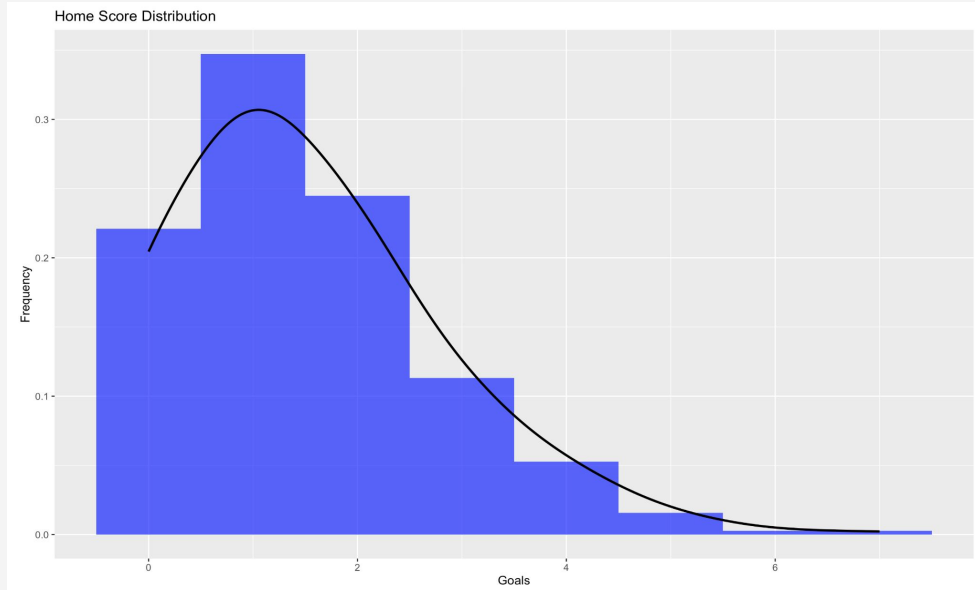
**W = .88365**

**P -VALUE = 2.325E-16**

**REJECT THE NULL HOME SCORE  
IS NOT NORMALLY  
DISTRIBUTED**



# **HOME SCORE DISTRIBUTION**

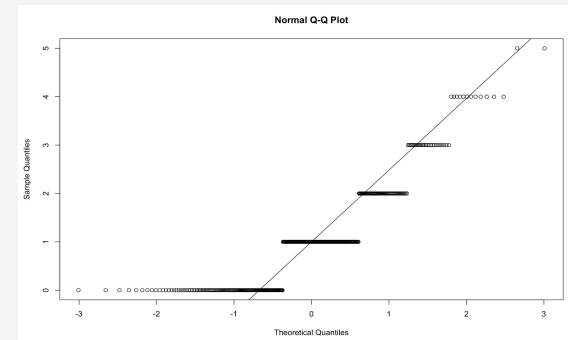


## **SHAPIRO-WILKS TEST**

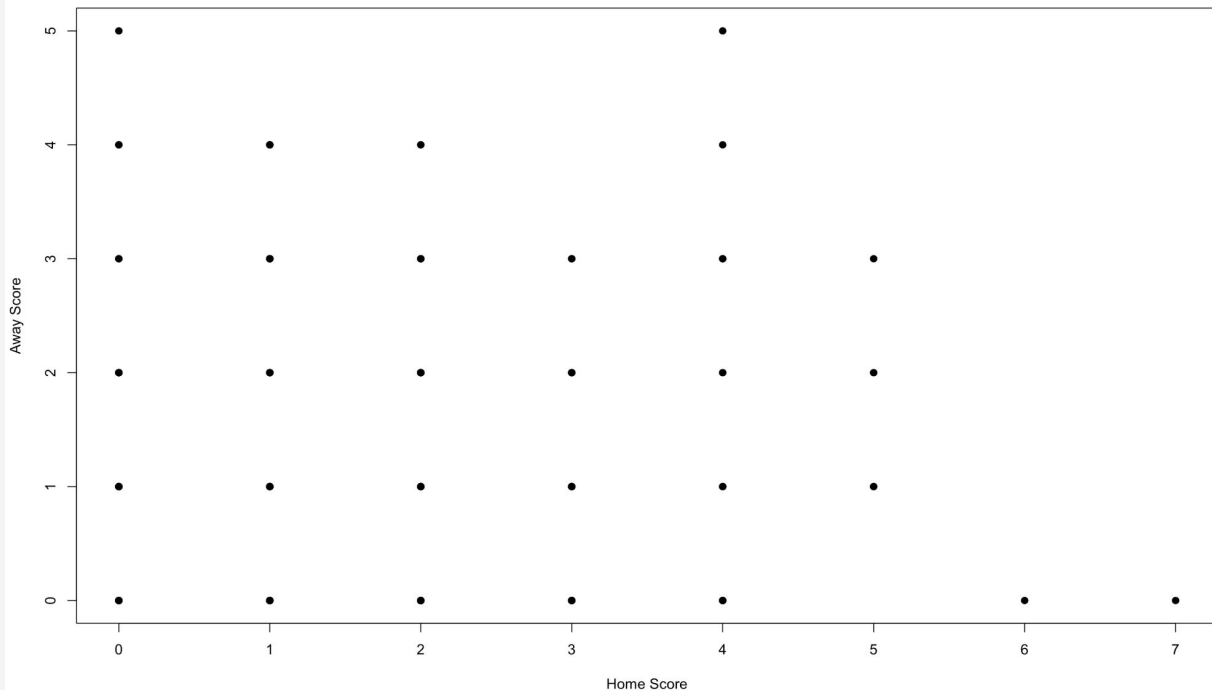
**W = .83144**

**P -VALUE = <2.2E-16**

**REJECT THE NULL AWAY SCORE  
IS NOT NORMALLY  
DISTRIBUTED**

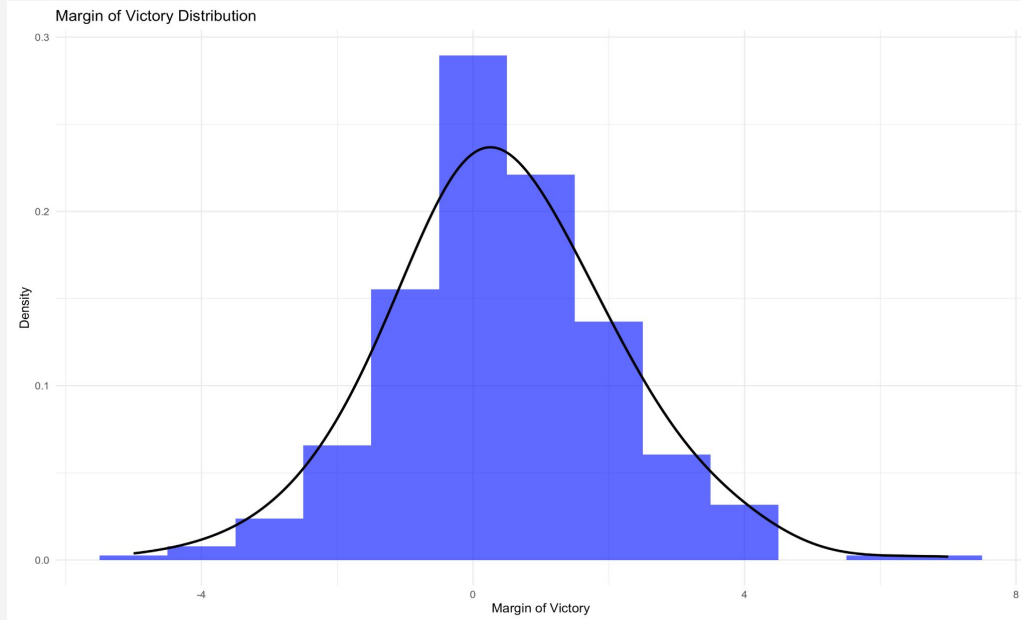


# **AWAY SCORE DISTRIBUTION**



**CORRELATION**  
**.0166054**

**SCATTER PLOT BETWEEN HOME + AWAY SCORE**

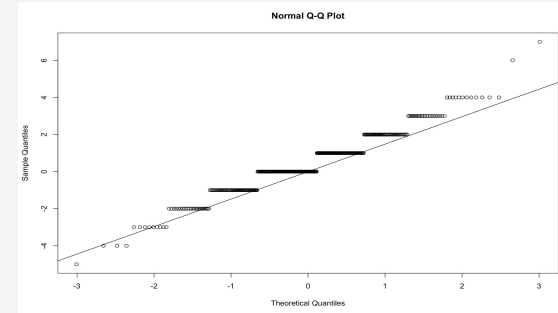


## **SHAPIRO-WILKS TEST**

**W = .96051**

**P-VALUE = 1.385E-08**

**REJECT THE NULL MARGIN OF  
VICTORY IS NOT NORMALLY  
DISTRIBUTED**



# MARGIN OF VICTORY

# DISCUSSION OF THE SCORING DISTRIBUTIONS

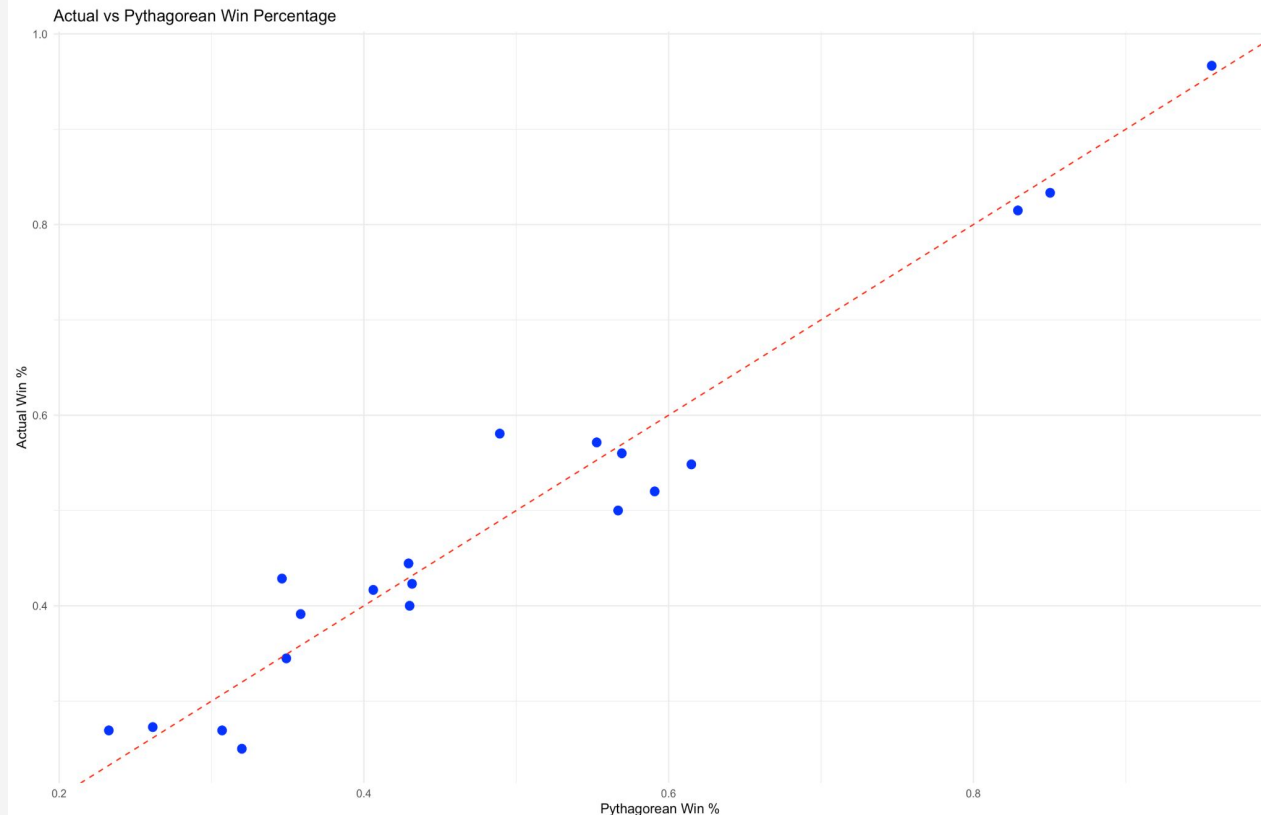
Since the scoring distributions are both not normal based off the Shapiro Wilks Test. The normality plots show a similar behavior. We must use the Poisson rating system later. The histograms both show a right skew implying that EPL soccer matches in the 2004-05 season had low scores. The home score had a slightly higher mean of 1.5 compared to the away score mean of 1.066 but both shared the same median of 1 goal. Home score also had a larger max number of goals scored in a match at 7 while away was smaller at 5 goals. They both had small standard deviations with home score continuing to be slightly higher at 1.25 while away score had a standard deviation of 1.08. The correlation between home score and away score is 0.0166 implying there is no linear relationship between the two variables. The margin of victory also displays significant variability, implying a competitive league structure. This analysis justifies the use of Poisson models in subsequent evaluations specifically Poisson ratings for offensive and defensive performance.

# PYTHAGOREAN WIN PERCENTAGES

|                   | Actual Win Percentage | Pythagorean Win Percentage | Error        |
|-------------------|-----------------------|----------------------------|--------------|
| Arsenal FC        | 0.8333333             | 0.8504021                  | -0.017068734 |
| Aston Villa       | 0.4444444             | 0.4292933                  | 0.015151162  |
| Birmingham City   | 0.4230769             | 0.4316201                  | -0.008543203 |
| Blackburn Rovers  | 0.3913043             | 0.3584999                  | 0.032804407  |
| Bolton Wanderers  | 0.5714286             | 0.5527939                  | 0.018634719  |
| Charlton Athletic | 0.4285714             | 0.3462276                  | 0.082343821  |
| Chelsea FC        | 0.9666667             | 0.956445                   | 0.010221699  |
| Crystal Palace    | 0.2692308             | 0.3069408                  | -0.037710012 |
| Everton FC        | 0.5806452             | 0.4891805                  | 0.091464667  |
| Fulham FC         | 0.4                   | 0.4300077                  | -0.030007713 |

|                      | Actual Win Percentage | Pythagorean Win Percentage | Error        |
|----------------------|-----------------------|----------------------------|--------------|
| Liverpool FC         | 0.5483871             | 0.6149258                  | -0.066538656 |
| Manchester City      | 0.52                  | 0.5908447                  | -0.070844675 |
| Manchester United    | 0.8148148             | 0.829222                   | -0.014407214 |
| Middlesbrough FC     | 0.56                  | 0.5692921                  | -0.009292079 |
| Newcastle United     | 0.4166667             | 0.4061509                  | 0.010515717  |
| Norwich City         | 0.2692308             | 0.2325959                  | 0.036634821  |
| Portsmouth FC        | 0.3448276             | 0.3491022                  | -0.004274634 |
| Southampton FC       | 0.25                  | 0.3198945                  | -0.069894519 |
| Tottenham Hotspur    | 0.5                   | 0.5668397                  | -0.066839749 |
| West Bromwich Albion | 0.2727273             | 0.2614277                  | 0.011299584  |

**CORRELATION**  
**.973436**



**PYTHAGOREAN WIN % VS. ACTUAL WIN %**

# OVER + UNDER ACHIEVERS

## **OVERACHIEVING TEAMS**

**THESE TEAMS HAVE THE LARGEST POSITIVE ERRORS MEANING THEY HAD A HIGHER ACTUAL WIN PERCENTAGE THAN THEIR PYTHAGOREAN WIN PERCENTAGE.**

**EVERTON (0.0915)  
CHARLTON ATHLETIC (.0823)  
NORWICH (.0366)  
BLACKBURN ROVERS (.0328)**

## **UNDERACHIEVING TEAMS**

**THESE TEAMS HAVE THE SMALLEST NEGATIVE ERRORS MEANING THEY HAD A LOWER ACTUAL WIN PERCENTAGE THAN THEIR PYTHAGOREAN WIN PERCENTAGE.**

**MANCHESTER CITY (-0.0708)  
SOUTHAMPTON (-0.0699)  
TOTTENHAM HOTSPUR (-0.0668)  
LIVERPOOL (-0.0665)**

# DISCUSSION OF PYTHAGOREAN WIN PERCENTAGES

The Pythagorean win percentage pretty accurately captures team performance and the strong positive correlation between the two variables. Even the largest overachievers and underachievers don't stray far. The largest errors are smaller than 10% (Everton at 9.15%) and the average error was less than half a percent (.43%).

# POISSON RATINGS

| Team              | Offense     | Defense      |
|-------------------|-------------|--------------|
| Arsenal FC        | 0.84381974  | -0.487076061 |
| Aston Villa       | 0.19996382  | -0.164320865 |
| Birmingham City   | 0.07543473  | -0.29260374  |
| Blackburn Rovers  | -0.15130034 | -0.368672897 |
| Bolton Wanderers  | 0.27669303  | -0.3274873   |
| Charlton Athletic | 0.13729324  | -0.058072779 |
| Chelsea FC        | 0.63116299  | -1.380935036 |
| Crystal Palace    | 0.11748417  | 0.007725309  |
| Everton FC        | 0.1934821   | -0.287237476 |
| Fulham FC         | 0.35376598  | -0.013097983 |
| Liverpool FC      | 0.33302764  | -0.395043919 |

| Team                 | Offense    | Defense      |
|----------------------|------------|--------------|
| Manchester City      | 0.22955217 | -0.450540782 |
| Manchester United    | 0.42636356 | -0.845029265 |
| Middlesbrough FC     | 0.3575409  | -0.278584772 |
| Newcastle United     | 0.24902151 | -0.070063973 |
| Norwich City         | 0.15820279 | 0.226252835  |
| Portsmouth FC        | 0.16198222 | -0.039838783 |
| Southampton FC       | 0.21527062 | 0.074840899  |
| Tottenham Hotspur    | 0.23169484 | -0.400422705 |
| West Bromwich Albion | NA         | NA           |
| Manchester City      | 0.22955217 | -0.450540782 |
| Manchester United    | 0.42636356 | -0.845029265 |

# TEAM OFFENSIVE QUALITY BASED OFF POISSON RATINGS

## WORST TEAMS

- **BLACKBURN ROVERS (-.1513)**
- **BIRMINGHAM CITY (.0754)**
- **CRYSTAL PALACE (.1175)**

## AVERAGE TEAMS

- **WEST BROMWICH ALBION (BASELINE)**
- **LIVERPOOL (.3330)**

## BEST TEAMS

- **ARSENAL (.8438)**
- **CHELSEA (.6312)**
- **MANCHESTER UNITED (.4264)**

The baseline for both offense and defense are the West Bromwich Albion. Arsenal, Chelsea and Manchester United being the best offensively aligns with them all finishing in the top three and scoring the most goals across the Premier League. While Crystal Palace was relegated. Rovers and Birmingham City also had a poor performance offensively with all three teams having the lowest number of goals scored.

# TEAM DEFENSIVE QUALITY BASED OFF POISSON RATINGS

## WORST TEAMS

- **NORWICH CITY (.2263)**
- **SOUTHAMPTON (.0748)**
- **CRYSTAL PALACE (.0077)**

## AVERAGE TEAMS

- **WEST BROMWICH ALBION (BASELINE)**
- **BOLTON WANDERERS (-.3274)**

## BEST TEAMS

- **CHELSEA (-1.3809)**
- **MANCHESTER UNITED (-.8450)**
- **ARSENAL (-.4871)**

Arsenal, Chelsea and Manchester United being the best defensively also aligns with them all finishing in the top three and conceding the least goals across the Premier League. Crystal Palace being one of the worst teams offensively and defensively but just barely getting relegated is a very interesting. All the worst defensive teams were relegated.

# ELO RATINGS

## HOME FIELD ADVANTAGE

**HFA: 76.04813**

## WEIGHTING FACTOR (K)

**K-VALUE: 5**

|                   | Starting Elo | Ending Elo | Difference |
|-------------------|--------------|------------|------------|
| Arsenal FC        | 1500         | 1532.919   | 32.919411  |
| Aston Villa       | 1500         | 1485.449   | -14.550558 |
| Birmingham City   | 1400         | 1406.33    | 6.32957    |
| Blackburn Rovers  | 1500         | 1482.676   | -17.324285 |
| Bolton Wanderers  | 1400         | 1422.673   | 22.673198  |
| Charlton Athletic | 1400         | 1405.149   | 5.148851   |
| Chelsea FC        | 1500         | 1549.867   | 49.867105  |
| Crystal Palace    | 1500         | 1468.222   | -31.778183 |
| Everton FC        | 1500         | 1501.184   | 1.184192   |
| Fulham FC         | 1400         | 1402.612   | 2.611998   |
| Liverpool FC      | 1500         | 1498.072   | -1.927838  |

|                      | Starting Elo | Ending Elo | Difference |
|----------------------|--------------|------------|------------|
| Manchester City      | 1500         | 1495.192   | -4.80818   |
| Manchester United    | 1500         | 1526.915   | 26.914645  |
| Middlesbrough FC     | 1500         | 1497.979   | -2.021051  |
| Newcastle United     | 1400         | 1405.9     | 5.900498   |
| Norwich City         | 1500         | 1468.259   | -31.741067 |
| Portsmouth FC        | 1400         | 1395.205   | -4.794542  |
| Southampton FC       | 1500         | 1468.354   | -31.645857 |
| Tottenham Hotspur    | 1500         | 1492.253   | -7.74704   |
| West Bromwich Albion | 1400         | 1394.789   | -5.210865  |
| Manchester City      | 1500         | 1495.192   | -4.80818   |
| Manchester United    | 1500         | 1526.915   | 26.914645  |

# ELO RATINGS - IMPROVEMENTS AND DECLINES

## DECLINING TEAMS

- **NORWICH CITY (-31.74)**
- **SOUTHAMPTON FC (-31.65)**
- **CRYSTAL PALACE (-31.78)**
- **BLACKBURN ROVERS (-17.32)**

## CONSISTENT TEAMS

- **EVERTON FC (+1.18)**
- **LIVERPOOL FC (-1.93)**

## IMPROVING TEAMS

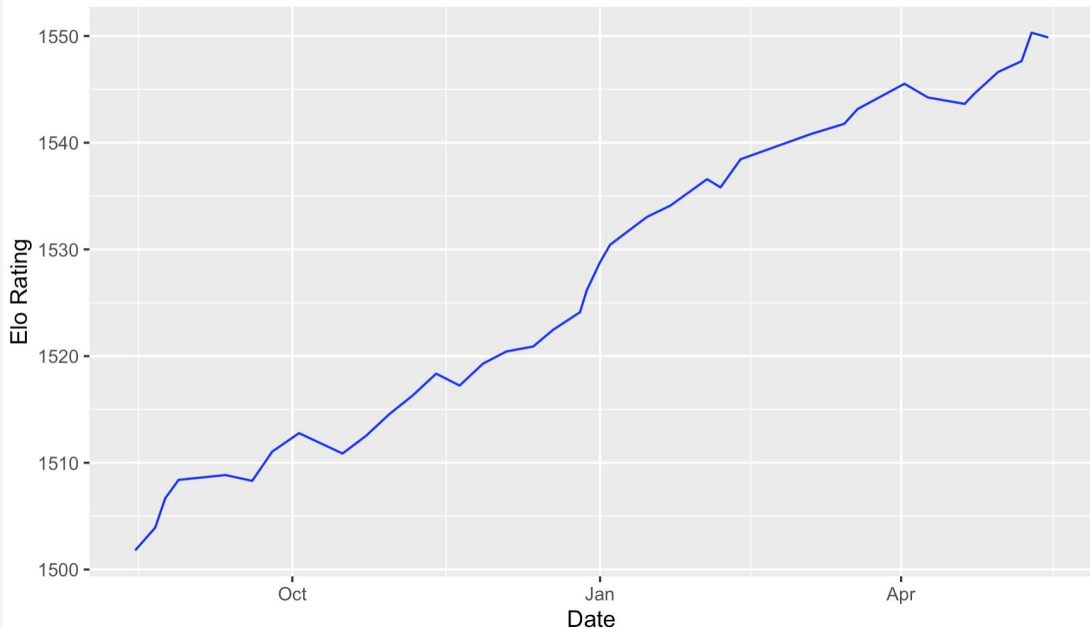
- **CHELSEA FC(+49.87)**
- **ARSENAL FC (+32.92)**
- **MANCHESTER UNITED (+26.91)**
- **BOLTON WANDERERS (+22.67)**

The Elo ratings effectively capture team trajectories throughout the season after they regress towards the mean from the previous season. Chelsea and Arsenal with the largest Elo improvements led were driven by consistent victories and dominant performances. All the teams with the largest Elo improvements bar Bolton finished in the top 4 and Bolton was only a win away from top 4.

The three out of the four declining teams were relegated. Norwich City, Southampton and Crystal Palace all had similar Elo declines as well which makes sense because the three teams all finished within a point of each other. However the Blackburn Rovers had a ten point cushion between themselves and the relegation teams.

# CHELSEA FC ELO RATING

Elo Progression for Chelsea FC



Chelsea's Elo progression (+49.87) underscores their dominance during the 2004-05 season. This improvement was driven by their record-breaking defense and consistent performances under new management. Even though Chelsea finished second the previous season Chelsea gained an additional 16 points in 2004-05. This is demonstrated by the relatively linear growth of Chelsea's Elo progression.

# SIMULATED SEASON RESULTS

## AFTER 5000 ITERATIONS

| Team              | Actual Points | Average Simulated Points | Error    | EPL Titles |
|-------------------|---------------|--------------------------|----------|------------|
| Arsenal FC        | 83            | 55.4136                  | 27.5864  | 0.1724     |
| Aston Villa       | 47            | 51.1054                  | -4.1054  | 0.0416     |
| Birmingham City   | 45            | 44.6838                  | 0.3162   | 0.0016     |
| Blackburn Rovers  | 42            | 50.9122                  | -8.9122  | 0.0374     |
| Bolton Wanderers  | 58            | 46.2502                  | 11.7498  | 0.0018     |
| Charlton Athletic | 46            | 44.6494                  | 1.3506   | 0.0012     |
| Chelsea FC        | 95            | 56.898                   | 38.102   | 0.2626     |
| Crystal Palace    | 33            | 49.7092                  | -16.7092 | 0.025      |
| Everton FC        | 61            | 52.3924                  | 8.6076   | 0.0578     |
| Fulham FC         | 44            | 44.4524                  | -0.4524  | 0.0006     |

| Team                 | Actual Points | Average Simulated Points | Error    | EPL Titles |
|----------------------|---------------|--------------------------|----------|------------|
| Liverpool FC         | 58            | 52.1528                  | 5.8472   | 0.0504     |
| Manchester City      | 52            | 51.9404                  | 0.0596   | 0.0608     |
| Manchester United    | 77            | 54.8646                  | 22.1354  | 0.1228     |
| Middlesbrough FC     | 55            | 52.2266                  | 2.7734   | 0.07       |
| Newcastle United     | 44            | 44.6666                  | -0.6666  | 0.001      |
| Norwich City         | 33            | 49.7306                  | -16.7306 | 0.0278     |
| Portsmouth FC        | 39            | 43.8926                  | -4.8926  | 0.0008     |
| Southampton FC       | 32            | 49.742                   | -17.742  | 0.0232     |
| Tottenham Hotspur    | 52            | 51.6358                  | 0.3642   | 0.0406     |
| West Bromwich Albion | 34            | 43.8214                  | -9.8214  | 0.0006     |

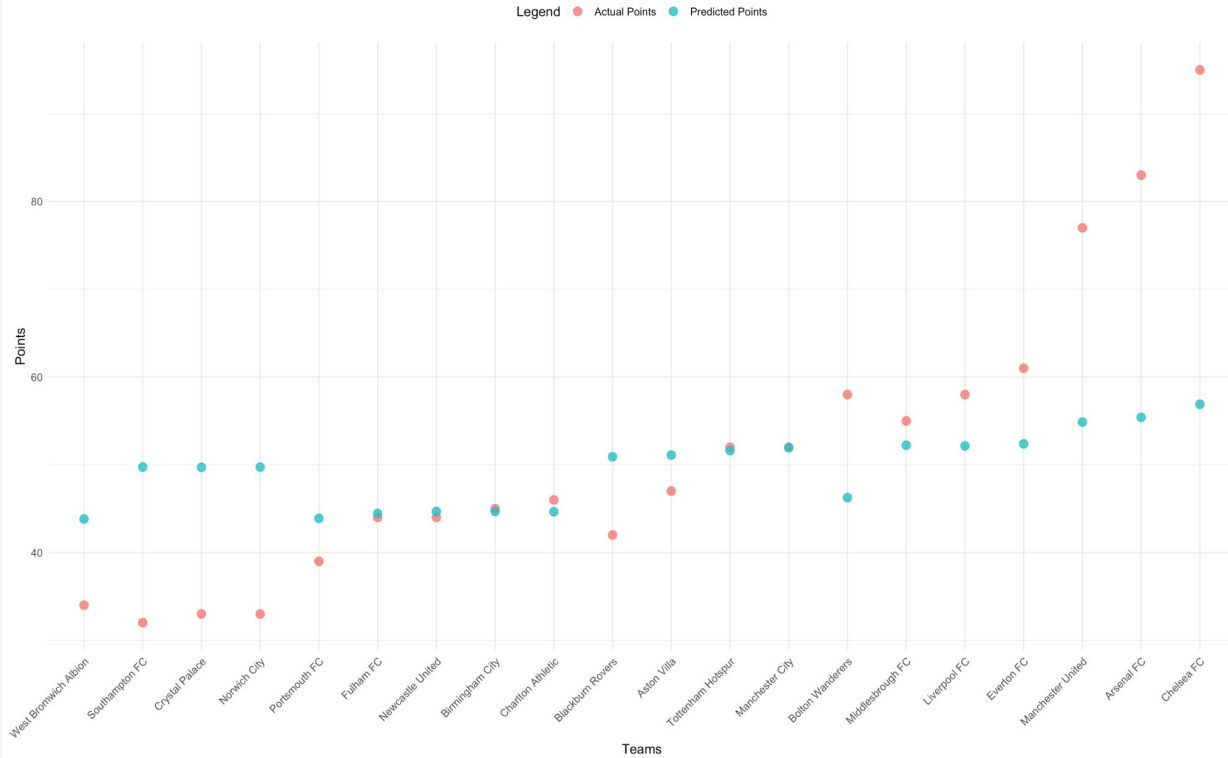
# PREDICTION TO REALITY

## TAKING SIMULATED AVERAGE POINTS AND CREATING EPL STANDINGS

| Finish | Simulated Season EPL Standings | Actual EPL Standings | Position Change |
|--------|--------------------------------|----------------------|-----------------|
| 1      | Chelsea FC                     | Chelsea FC           | 0               |
| 2      | Arsenal FC                     | Arsenal FC           | 0               |
| 3      | Manchester United              | Manchester United    | 0               |
| 4      | Everton FC                     | Everton FC           | 0               |
| 5      | Middlesbrough FC               | Liverpool FC         | -2              |
| 6      | Liverpool FC                   | Bolton Wanderers     | +1              |
| 7      | Manchester City                | Middlesbrough FC     | -1              |
| 8      | Tottenham Hotspur              | Manchester City      | -1              |
| 9      | Aston Villa                    | Tottenham Hotspur    | -1              |
| 10     | Blackburn Rovers               | Aston Villa          | -5              |

| Finish | Simulated Season EPL Standings | Actual EPL Standings | Position Change |
|--------|--------------------------------|----------------------|-----------------|
| 11     | Southampton FC                 | Charlton Athletic    | -9              |
| 12     | Norwich City                   | Birmingham City      | -7              |
| 13     | Crystal Palace                 | Fulham FC            | -5              |
| 14     | Bolton Wanderers               | Newcastle United     | +8              |
| 15     | Birmingham City                | Blackburn Rovers     | +3              |
| 16     | Newcastle United               | Portsmouth FC        | +2              |
| 17     | Charlton Athletic              | West Bromwich Albion | +6              |
| 18     | Fulham FC                      | Crystal Palace       | +5              |
| 19     | Portsmouth FC                  | Norwich City         | +3              |
| 20     | West Bromwich Albion           | Southampton FC       | +3              |

Actual Points vs Predicted Points in Monte Carlo Simulation



**CORRELATION**  
**.705043**

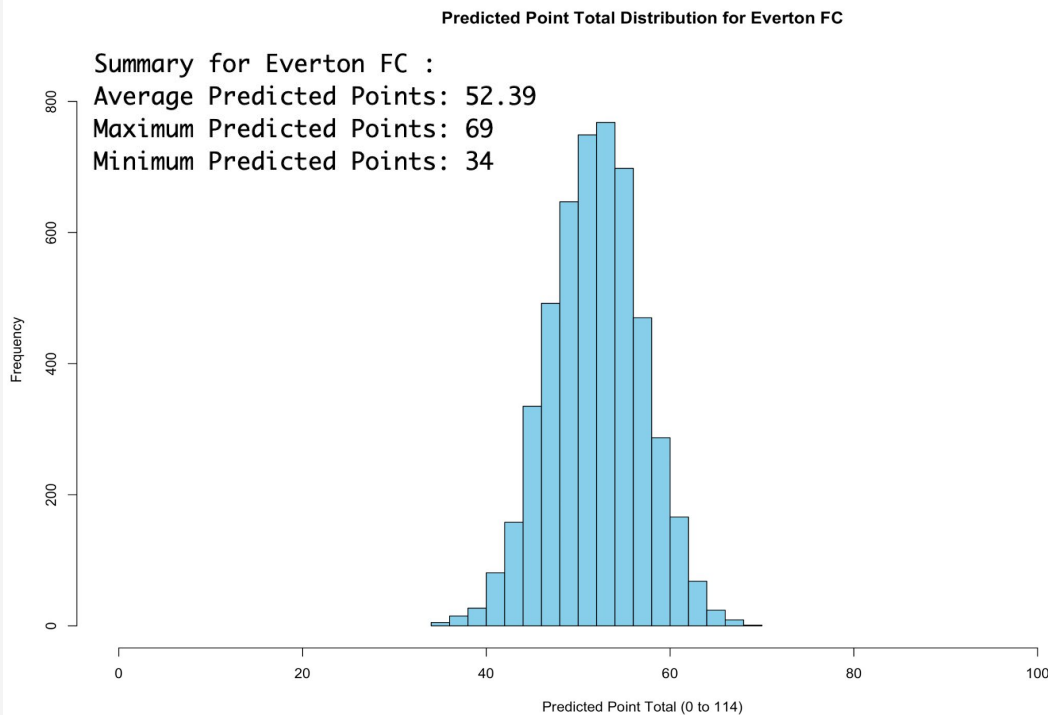
**SIMULATED SEASON VS. ACTUAL SEASON**

# DISCUSSION OF SIMULATED SEASON VS. ACTUAL SEASON

The Monte Carlo simulations achieved a strong correlation (0.705) with actual standings, showing that there is a decent linear relationship and demonstrating the model's accuracy. The model seemed to prefer a more competitive table with no team having a larger point total than around 57 points, which was Chelsea. In reality, Chelsea finished with 95 points which caused wide errors for the top and bottom half of the league.

Taking the average simulated points I created a standings table and compared it to the actual results. It was strong at predicting the top 9 with only slight deviations between the results but had large deviations from the results for the bottom half. The top half the worst deviation was 5 places off for predicting Blackburn Rovers would finish in 10th instead of 15th while the worst deviation in the bottom half was 9 places for predicting Southampton to finish in 11th when they finished at the very bottom of the table. These differences highlight the inherent unpredictability of the Premier League, influenced by external factors such as injuries, form, and managerial decisions, which simulations cannot fully capture. Additionally, the difference between 7th to 13th and from 14th to 20th is less than three points (2.23 point difference from 7 to 12 and 2.43 point difference between 14 to 20). While the difference between 1st to 6th is wider at 4.74 points. These smaller differences between the bottom half may attribute to the larger deviations.

# ANALYSIS OF AN OVERACHIEVER - EVERTON FC



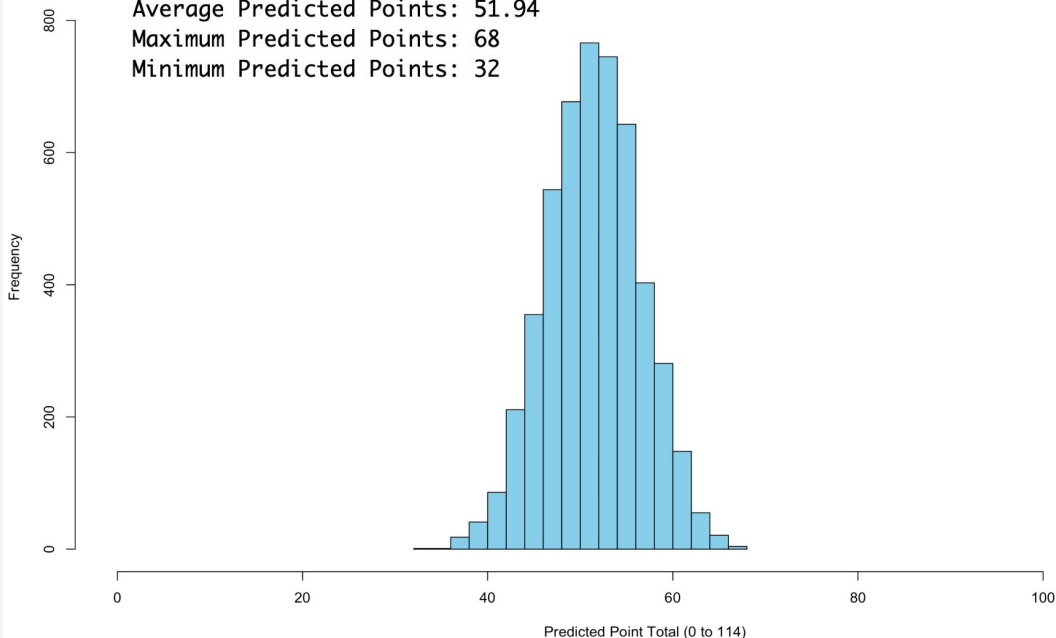
On average Everton finished with around 53 points. Their actual point total of 61 points is on the extreme right side of the tail demonstrating their overperformance. This could be achieved by their strong defense since they had an Elo rating in the top half. Additionally, luck and other in game dynamics may have influenced Everton's overperformance. It is also important to point out that Everton while the largest overachiever only overachieved by 9% from their actual win percentage.

The histogram appears relatively normal. The Monte Carlo Simulation aligns with the Pythagorean Win Percentage by predicting a worse performance than their actual performance.

# ANALYSIS OF AN UNDERACHIEVER - MAN CITY

Predicted Point Total Distribution for Manchester City

Summary for Manchester City :  
Average Predicted Points: 51.94  
Maximum Predicted Points: 68  
Minimum Predicted Points: 32



Manchester City underachieved by around 7%. Their simulated point performance aligned with their actual point performance (51.94 versus 52 points). Based off their Poisson ratings Manchester City should have a moderately strong defense and offense. It appears Manchester City should have been a stronger team but failed to achieve the heights they should've. 10 out of the 12 losses Manchester City had this season had a one goal margin of victory which may have contributed to them underperforming.

The histogram appears relatively normal with slight left skew. The Monte Carlo Simulation seems to contradict the pythagorean win percentages by accurately predicting Manchester City's season.

# KEY POINTS

## Relevant Statistics

- Strong correlation between goals/shots and points per match.
- Home teams scored more goals on average than away teams.

## Scoring Distributions

- Non-normal, right-skewed distributions. Poisson models needed for ratings.

## Pythagorean Win Percentage

- Pythagorean win percentage very strong linear relationship.
- Everton (Overachiever: +9.15% error).
- Manchester City (Underachiever: -7.08% error).

## Poisson Ratings

- Best Offense: Arsenal, Chelsea, Manchester United.
- Best Defense: Chelsea, Manchester United.

## Elo Ratings

- Two most improved teams finished in the top two and three of the most declined teams were relegated.

## Monte Carlo Simulation

- Chelsea predicted as EPL winners in 26% of simulations.
- Strong top-half predictions but struggles in the bottom half of the table.

## Under & Over Achievers

- Everton: Strong results relative to Pythagorean and simulated points.
- Manchester City: Points far below simulation expectations.

# REFERENCES

Premier League. (n.d.). *Season reviews: 2013/14*.

<https://www.premierleague.com/history/season-reviews/13>



**THANKS!**

